But from table of area under Standard Normal curve

$P(0 < Z < 1.28) = 0.3997$ ... (2)

$\therefore$ From (1) and (2)

$$\frac{4}{\sigma} = 1.28$$

$$\sigma = 3.125$$

(ii) The number of plants with height greater than 105 cms if there were 200 plants in the garden.

$$X = 105$$

$$Z = \frac{x-\mu}{\sigma} = \frac{105-100}{3.125}$$

$$= \frac{5}{3.125}$$

$$= 1.6$$

$P(1.6 < Z < \infty) = 0.5\text{-}0.4452 = 0.0548$

Number of plants $= 200 \times 0.0548 = 10.9600 \simeq 11$

$\therefore$ 11 plants have height greater than 105cms.

Example 10.31

Students of a class were given an aptitude test. Their marks were found to be normally distributed with mean 60 and standard deviation 5. What percentage of students scored (i) more than 60 marks (ii) less than 56 marks (iii) between 45 and 65 marks.

Solution:

Given mean $\mu = 60$ and standard deviation $\sigma = 5$

Standard normal variate $Z = \frac{x-\mu}{\sigma} = \frac{x-60}{5}$

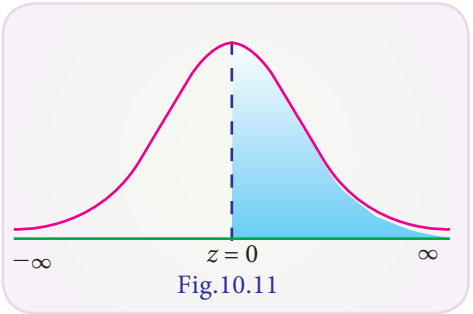


Fig.10.11

(i) P (more than 60) $= P(x > 60)$

$$= P\left(z > \frac{60-60}{5}\right)$$

$$= P(Z > 0)$$

$$= P(0 < Z < \infty)$$

$$= 0.5000$$

Student scored more than 60 marks = 0.5000 x 100

$$= 50\%$$